

Model & Data Card — TrafficLens (ATIP)

Team KUET_Technomancers · SciBlitz AI Challenge 2026 · IEEE SB CUET

Pre-trained models used

Model	Provider / access	Role	License
YOLO11n (yolo11n.pt)	Ultralytics (local inference)	Vehicle detection	AGPL-3.0
BoT-SORT tracker	Ultralytics implementation	Multi-object tracking	AGPL-3.0 (via Ultralytics)
Llama-4-Scout-17B-16E-Ins truct	Meta, served via Groq API	Vision-language vehicle description	Llama 4 Community License
Llama-3.3-70B-Versatile	Meta, served via Groq API	Query parsing & semantic re-ranking	Llama 3.3 Community License

Datasets

Dataset	Source	Use in this project	License
COCO (via pretrained weights)	cocodataset.org — used only implicitly, as YOLO11n ships pretrained on COCO	We performed no training or fine-tuning; detection classes car/bus/truck/motorcycle come from COCO	Annotations CC BY 4.0
User-supplied traffic videos	Uploaded by the operator (test clips: publicly available traffic-camera footage)	Inference-time input only; never used for training; deleted with the video	Uploader's responsibility

Key third-party libraries

PyTorch (BSD-3), Ultralytics (AGPL-3.0), OpenCV (Apache-2.0), Streamlit (Apache-2.0), Groq SDK (Apache-2.0), pandas (BSD-3), Plotly (MIT), NumPy (BSD-3), lapx (BSD-2), FFmpeg (LGPL/GPL), SQLite (public domain). Full list in requirements.txt; all are used unmodified with attribution.

Known limitations

- Attribute quality degrades with small/blurred/occluded crops; plate and text transcription may be partial or wrong — outputs are investigative leads, not evidence.
- VLM outputs can hallucinate details despite prompt constraints; dominant-color-only matching and empty-list rules reduce but do not eliminate this.
- Pretrained COCO detection may underperform on region-specific vehicles (e.g., rickshaws, CNGs) not in its class set.
- No persistent storage on the free deployment tier; data resets on restart.

Ethical considerations

- Surveillance context: the tool describes vehicles, not people; we do not perform face detection/recognition or person identification.
- Misidentification risk: watchlist and plate matches are clearly presented with evidence crops for human verification; the system is decision-support, not automated enforcement.
- Privacy: uploaded footage stays in the processing container, is not used for training, and can be deleted by the operator in one click; API keys are stored as server-side secrets.
- Bias: detection/description quality may vary across vehicle types and lighting conditions common in different regions; operators should validate on local footage before relying on it.